

To begin unifying the two tracks, we asked whether the reasoning chain and final answer require different amounts of update information. To test this hypothesis, we fine-tuned rank-16 all-linear LoRA adapters on MetaMathQA with two frozen NF4 backbones, Mistral-7B and Qwen2.5-7B. For each model, we trained on the full response, the reasoning span only, or the answer span only.

After training, each fixed adapter was compressed through the same dense codec ladder used in [Section 3](#). We measured the exact serialized file rate in bits per LoRA value and the signed held-out bits per token saved against the frozen model, separately for the reasoning and answer spans. For a frozen model M and span utility U , we write

$$B_{\text{span}}^*(\tau) = \min B(A) \quad \text{such that} \quad W_{MU}(A) \geq \tau W_{MU}(A_{\text{full}}) \quad (4)$$